DS 303: Introduction to Machine Learning Sample Questions 1

Prepared by Soumen Mondal, Sneha Oram

January 25, 2026

1 Logistic Regression with Regularization [6 points]

Consider a discriminative classifier for solving the binary classification problem as illustrated in Figure 1.

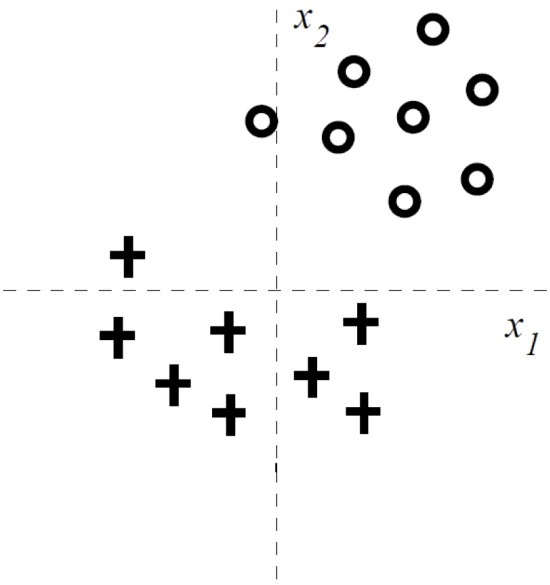

Figure 1: The training set with 2 features (x_1, x_2), where '+' corresponds to class $y = 1$ and 'o' corresponds to class $y = 0$.

Consider solving the binary classification task depicted in Figure 1 with the simple linear logistic regression model as follows:

$$P(y = 1|\mathbf{x}, \mathbf{w}, w_0) = \sigma(\mathbf{w}^T\mathbf{x} + w_0) \tag{1}$$

Where $\mathbf{w} = (w_1, w_2)$, $\mathbf{x} = (x_1, x_2)$ and $\sigma(.)$ is the sigmoid function.

(a) [**3 points**] Consider training objective of regularized linear logistic regression models as follows:

$$\max_{\mathbf{w}, w_0} \sum_{i=1}^{N} \log(P(y_i|x_i, \mathbf{w}, w_0)) - C \cdot |w_j| \tag{2}$$

Note that here C is very large and only one of the parameter w_j is regularized in the objective function instead of the L_1 norm of parameters. From Figure 1, it is clear that the training

data can be separated with zero training error with a linear separator. Given the training data in Figure 1, how does the training error change with regularization of each parameter w_j where $j \in \{0, 1, 2\}$? State whether the training error increases or stays the same (zero) with justification. (No credit if justification is not provided)

(b) [**1 points**] If we change the form of regularization to L_2 norm and regularize w_1 and w_2 only (but not w_0), we get the following training objective:

$$\max_{\mathbf{w}, w_0} \sum_{i=1}^{N} \log(P(y_i|x_i, \mathbf{w}, w_0)) - C \cdot \sqrt{(|w_1|^2 + |w_2|^2)} \tag{3}$$

If we increase the regularization parameter C which of the following scenarios do you expect to observe? Provide justification for your choice. (Contains negative credit for incorrect option. No credit if justification is not provided)

1. First w_1 will become 0, then w_2.
2. First w_2 will become 0, then w_1.
3. w_1 and w_2 will become zero simultaneously.
4. None of the weights will become exactly zero.

(c) [**1 points**] For very large C, with the same L_2 norm regularization for w_1 and w_2 as above, which value(s) do you expect w_0 to take if the number of points in each class is same?

(d) [**1 points**] Assume that we obtain more data points from the '+' class that corresponds to $y = 1$ so that the class labels become unbalanced. Again for very large C, with the same L_2 norm regularization for w_1 and w_2 as above, which value(s) do you expect w_0 to take?

2 Linear Regression with Gaussian Noise [4 points]

Consider a linear model with D features and it's loss function as follows:

$$y(\mathbf{x}, \mathbf{w}, w_0) = w_0 + \mathbf{w}^T\mathbf{x} \tag{4}$$

$$L(\mathbf{w}, w_0) = \frac{1}{2}\sum_{j=1}^{N}[y(\mathbf{x}^{(j)}, \mathbf{w}, w_0) - y_{true}]^2 \tag{5}$$

Now suppose that Gaussian noise ϵ_i with $E[\epsilon_i] = 0$ and $Var[\epsilon_i] = 1$ is added independently to each of the input features $x_i^{(j)}$. Moreover, $E[\epsilon_p \cdot \epsilon_q] = \delta_{pq}$ (delta function). Show that the expectation of loss function L over the noise distribution, ϵ is equivalent to the sum-of-squares error for noise-free input variables, $\mathbf{x}$ with the addition of a weight-decay regularization term, in which the bias parameter w_0 is omitted from the regularizer. Equivalently, it is same as showing the following expression:

$$E_\epsilon[L(\mathbf{w}, w_0)] = \frac{1}{2}\sum_{j=1}^{N}[y(\mathbf{x}^{(j)}, \mathbf{w}, w_0) - y_{true}]^2 + \frac{1}{2}\sum_{i=1}^{D}|w_i|^2 \tag{6}$$

3 Error in Linear Regression [6 points]

Suppose we are in the year of 2026. The LIBS instrument of India's Chandrayaan-4 rover has made the first-ever in-situ measurements on the search of water ice beneath the lunar surface near the south pole of Moon. The instrument has just sent back exciting data giving the concentration of water ice y at depth x beneath the surface of the south pole on Moon! Your task, as one of the mission specialists (back on Earth), is to figure out what hypothesis best models the data, which is shown in Figure 2.

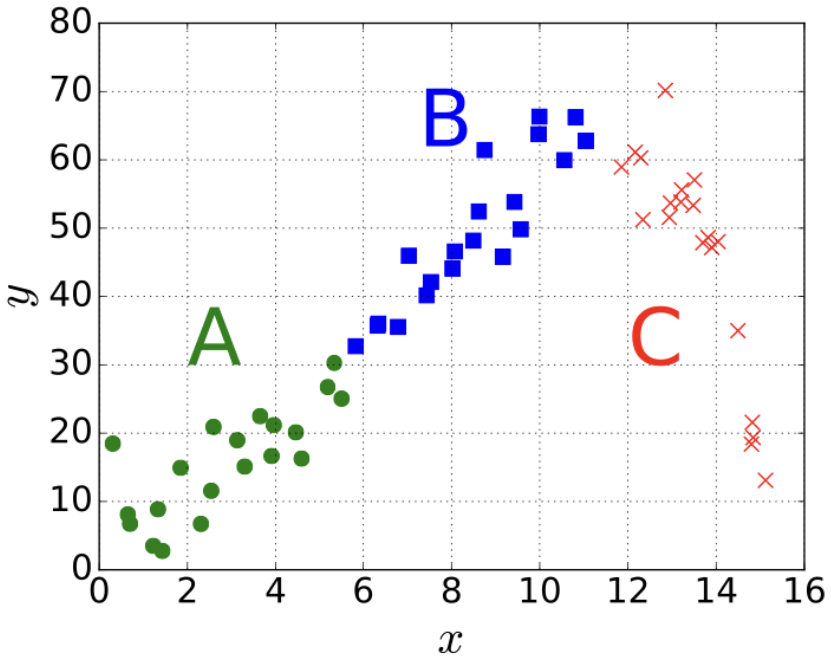

Figure 2: The concentration of water ice y at depth x beneath the surface of the south pole on Moon

Assume that the datapoints shown in this figure come from three disjoint subsets:

A: Depth $x = 0$ to $x = 6$ (circles)

B: Depth $x = 6$ to $x = 12$ (squares)

C: Depth $x > 12$ (the symbol $\times$)

And as an ML expert, you know that while you may train your model on one subset of data, you should test it on a different subset of data.

(a) [**2 points**] Suppose your hypothesis is that ice concentration is linearly related to depth, i.e. $y = \mathbf{w}^T\mathbf{x} + w_0$. You employ mean square error (MSE) for the objective function, and use dataset A for training, and dataset B for testing (since they are conveniently disjoint!). Let us say that that MSE below 30 is LOW, and MSE above 100 is HIGH. Judging from the above plot, will the MSE for training be LOW or HIGH? How about for testing? Explain why.

(b) [**2 points**] Continuing with the hypothesis that ice concentration is linearly related to depth, you now employ datasets A and B (combined) for training, and dataset C for testing.

Judging from the above plot, will the MSE for training be LOW or HIGH? How about for testing? Are your choices for training and testing datasets good ones? Explain.

(c) [**2 points**] Realizing that Moon is unlikely to be a snowball of ice (although it's possible Earth once was!), you switch to a family of hypotheses with nonlinear feature transforms, $y = \mathbf{w}^T\phi_k(\mathbf{x}) + w_0$, where $\phi_k(\mathbf{x})$ is a vector of polynomials up to order k. Can you think of any good way to evaluate what order k is the best to choose? Explain.

4 Loss Function of Logistic Regression [4 points]

For the binary logistic regression problem, the target values are encoded as $t^i \in \{0, +1\}$. For a dataset $D_N = \{(\mathbf{x}^{(i)}, t^{(i)})\}$ with $t^i \in \{0, +1\}$, the logistic regression is defined using the following steps:

$$z = \mathbf{w}^T\mathbf{x} + b \tag{7}$$

$$y = \sigma(z) \tag{8}$$

$$L(y, t) = -t\log(y) - (1-t)\log(1-y) \tag{9}$$

Show that if $t^i \in \{-1, +1\}$ then the minimization problem takes the following form where $\mathbf{w}$ and b are the weights parameters:

$$\min_{\mathbf{w},b} \sum_{i=1}^{N} \log(1 + \exp(-t^i(\mathbf{w}^T\mathbf{x}^{(i)} + b))) \tag{10}$$

5 Convexity of Sigmoid Function [3 points]

Prove that the following sigmoid function is convex, for any $k > 0$. Specify the region where the function is convex.

$$\sigma(kx) = \frac{1}{1 + e^{-kx}} \tag{11}$$

6 Coefficient of Determination [4 points]

Answer the following on coefficient of determination:

(a) [**2 points**] Explain how the unexplained sum of squares, and the total sum of squares are affected, when the model is poorly explaining an outlier.

(b) [**2 points**] Explain how does of the least square error differ from that of coefficient of determination in fitting the data?